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PII: S0021-9797(16)30867-0
DOI: <http://dx.doi.org/10.1016/j.jcis.2016.11.006>
Reference: YJCIS 21734

To appear in: *Journal of Colloid and Interface Science*

Received Date: 23 August 2016
Revised Date: 31 October 2016
Accepted Date: 2 November 2016



Please cite this article as: Z. Guo, D-d. Li, X-k. Luo, Y-h. Li, Q-N. Zhao, M-m. Li, Y-t. Zhao, T-s. Sun, C. Ma, Simultaneous determination of trace Cd(II), Pb(II) and Cu(II) by differential pulse anodic stripping voltammetry using a reduced graphene oxide-chitosan/poly-L-lysine nanocomposite modified glassy carbon electrode, *Journal of Colloid and Interface Science* (2016), doi: <http://dx.doi.org/10.1016/j.jcis.2016.11.006>

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Simultaneous determination of trace Cd(II), Pb(II) and Cu(II) by differential pulse anodic stripping voltammetry using a reduced graphene oxide-chitosan/poly-L-lysine nanocomposite modified glassy carbon electrode

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Abstract

The reduced graphene oxide (RGO) and Chitosan (CS) hybrid matrix RGO-CS were coated onto the glassy carbon electrode (GCE) surface, then, poly-L-lysine films (PLL) were prepared by electropolymerization with cyclic voltammetry (CV) method to prepare RGO-CS/PLL modified glassy carbon electrode (RGO-CS/PLL/GCE) for the simultaneous electrochemical determination of heavy metal ions Cd(II), Pb(II) and Cu(II). Combining the advantageous features of RGO and CS, RGO and CS are used together because the positively charged CS can interact with the negatively charged RGO to prevent their aggregation. Furthermore, CS has many amino groups along its macromolecular chains and possessed strongly reactive with metal ions. Moreover, PLL modified electrodes have good stability, excellent permselectivity, more active sites and strong adherence to electrode surface, which enhanced

electrocatalytic activity. The RGO-CS/PLL/GCE was characterized voltammetrically using redox couples ($\text{Fe}(\text{CN})_6^{3-/4-}$), complemented with electrochemical impedance spectroscopy (EIS). Differential pulse anodic stripping voltammetry (DPASV) has been used for the detection of $\text{Cd}(\text{II})$, $\text{Pb}(\text{II})$ and $\text{Cu}(\text{II})$. The detection limit of RGO-CS/PLL/GCE toward $\text{Cd}(\text{II})$, $\text{Pb}(\text{II})$ and $\text{Cu}(\text{II})$ is $0.01 \mu\text{g L}^{-1}$, $0.02 \mu\text{g L}^{-1}$ and $0.02 \mu\text{g L}^{-1}$, respectively. The electrochemical parameters that exert influence on deposition and stripping of metal ions, such as supporting electrolytes, pH value, deposition potential, and deposition time, were carefully studied.

KEYWORDS: Reduced graphene oxide; Chitosan; Poly-L-lysine films; Simultaneous detection; Heavy metals ions

1. INTRODUCTION

Rapid industrialization has led to an increased discharges wastewater containing toxic heavy metals ions (such as $\text{Cd}(\text{II})$, $\text{Pb}(\text{II})$ and $\text{Cu}(\text{II})$ and $\text{Hg}(\text{II})$), which have adverse effects on ecological environment and human health [1]. Although $\text{Cu}(\text{II})$ is essential for human health, its excess will cause health problems [2]. Water containing $\text{Cu}(\text{II})$ beyond the prescribed limit of 1.3 mg/L may causes lung cancer and liver damage in human being [3]. $\text{Pb}(\text{II})$, which commonly exists in industrial and agricultural wastewater, is severely hazardous to human beings and living things. Long-term drinking water containing high level of $\text{Pb}(\text{II})$ would cause serious disorders, such as kidney disease, nausea, coma,

convulsions, cancer, along with subtly negative effects on metabolism and intelligence [4]. Cd(II) is also utilized in many processes, including batteries and pigments. Its accumulation in human body causes negative effects such as blood pressure complications and kidney [5].

Due to serious hazardous effects of heavy metal ions on human health and toxicity in the environment, it is important to develop a simple and highly sensitive analytical method for assessing environmental pollutants to improve the quality of environment and human life. Several methods about the detection of Cd(II), Pb(II) and Cu(II) have been reported using atomic absorption spectrometry(AAS) [6-8], inductively coupled plasma mass spectrometry (ICP-MS) [9, 10], high performance liquid chromatography-inductively coupled plasma mass spectrometry [11], spectrophotometry [12], X-ray absorption spectroscopy [13], Surface enhanced Raman spectroscopy (SERS) [14]. However, most of the above-mentioned methods have some disadvantages, such as expensive, time consuming, bulky, complicate pretreatment, inconvenient for onsite analysis. Compared with the above methods, electrochemical techniques have attracted great interest in the detection of heavy metal ions because of their remarkable sensitivity, portability and low cost [1, 15, 16]. The traditional mercury film electrode and hanging mercury electrode have good reproducibility and high sensitivity, but they have great risk to human health and environment. So, much attention has been

focused on how to construct high performance electrode with low toxicity. Differential pulse anodic stripping voltammetry (DPASV) is a simple and sensitive electroanalytical method for the analysis trace concentrations of heavy metal ions [17-19]. Moreover, a distinct advantage of ASV over AAS method is its ability to detect several metal ions simultaneously. So, the working electrode plays a critical role in the stripping analyses. An ideal working electrode should have multiple functions, including effective preconcentration, selective reaction surface and good repeatability.

Graphene, which comprises two-dimensional nanostructures of sp²-bonded carbon atoms in a hexagonal configuration, is an excellent nanomaterials for electrochemistry owing to its good electrical conductivity, low cost, mechanical properties and high specific surface area [20-22]. These characteristics are propitious for facilitating electron transport and promoting mass transfer kinetics at the electrode surface. [23]. However, graphene tends to agglomerate or restack due to the strong π - π stacking and van der Waals interactions [24], which hinders its further applications. In order to conquer this problem and extend its applications, it is essential to modify graphene with functional materials to form hybrid materials with improved properties. In order to prevent the restacking of graphene, metal, metal oxide and conducting polymers are used to improve graphene distribution [24-26]. Those hybrid materials display

novel catalytic, magnetic, or optoelectronic properties.

It is well known that chemical functionalization can tailor the electronic solid state properties of carbon supports, which could be propitious for the improvement of catalyst performance. Chitosan (CS), a poly (D-glucosamine), is a natural polysaccharide with primary amino groups in the polymer chain. Due to the outstanding properties of Chitosan, such as nontoxic, biocompatibility, hydrophilic and antibacterial activity, it has been extensively used in various fields, such as biomaterials [27], textile [28] , drug delivery [29] and environmental protection [30]. CS has been extensively investigated for several decades for adsorption of heavy metal ions in water treatment because of its excellent quality, multiple functional groups, as well as its favorable price. CS is regarded as a promising adsorbent because it has both amine and hydroxyl groups that can remove the heavy metal ions by forming stable metal chelates. Several investigations have shown that the RGO-CS hybrid systems show enhanced electrochemical properties [31-33].

Polymer modified electrodes have been paid increasing attention in electrochemical analysis due to their more active sites, excellent permselectivity, good stability and strong adherence to electrode surface. Poly (L-lysine) (PLL) film can be easily formed on electrode surface by electropolymerization of L-lysine. PLL has been receiving much attention owing to its good biocompatibility, versatility and easiness of the

preparation, relatively good solubility in water, a flexible structure framework and a good deal of active amino groups [34, 35], which make them suitable for applications in electrochemical determination.

Recent advanced in new materials, especially the development of new nano-hybrid materials, have opened a new era of analytical techniques. By tailored functionalization and assembly, these materials can be easily assembled on electrode surface for the fabrication of sensing electrodes and the sensitive and selective detection of heavy metal ions [15, 36, 37]. In order to meet this requirement, the inducing of specific functional groups onto the electrode with the desired properties is particularly attractive. Combining the advantageous features of RGO and CS, RGO-CS nanocomposites are used together. There are plenty of amine groups on the CS which can strongly reactive with metal ions. In addition, the huge surface area of RGO makes it have a large adsorption capacity. Moreover, the combination of RGO-CS with PLL can provide excellent electrochemical platforms for target heavy metals sensing due to the combining of enlarged active surface area, strong adsorption capability and the specific complexing ability of the CS and PLL in analysis of environmental samples. To the best of our knowledge, RGO-CS/PLL used as sensing materials for the detection of heavy metal ions hasn't been reported. Herein, we introduce RGO-CS/PLL modified GCE for DPASV in simultaneous determination of Cd(II), Pb(II) and

Cu(II). The detection limit ($S/V=3$) for Cd(II), Pb(II) and Cu(II) at this sensor were $0.01 \mu\text{g L}^{-1}$, $0.02 \mu\text{g L}^{-1}$ and $0.02 \mu\text{g L}^{-1}$, respectively, which were lower than that of on the reported sensors for simultaneous determination of Cd(II), Pb(II) and Cu(II). In addition, the proposed sensor has been applied to simultaneous determination of Cd(II), Pb(II) and Cu(II) in wastewater samples with high selectivity.

2. Experimental

2.1 Reagents

Natural graphite powder (320 mesh, spectrum pure), Chitosan (CS, deacetylation degree was about 75-85%) and L-lysine were purchased from Aladdin Reagent Co. (Shanghai, China). Working solutions containing Cd(II), Pb(II) and Cu(II) were prepared by diluting the corresponding standard stock solutions that were purchased from Aladdin (Shanghai, China). A 0.1M HAc-NaAc buffer solution (pH 4.5) was prepared by mixing appropriate amounts of HAc and NaAc and served as a supporting electrolyte during the analysis of heavy metal ions. Other chemical were obtained from Beijing Chemical Reagent Factory (Beijing, China). All of the chemicals used were analytical-regent and twice-distilled water was used in all experiments.

2.2 Apparatus

The phases of the samples were examined by powder X-ray diffraction (XRD) analysis using a SIEMENS D5005 diffractometer with

Cu K α radiation at 40 Kv and 30 mA. FT-IR spectra (4000-400 cm⁻¹) were recorded using a Nicolet 20 FT-IR Spectrometer (Thermo Fisher Scientific Int., USA) to identify the functional groups and chemical bonding of the prepared materials. Scanning electron microscopy (SEM) images were obtained from FEI/Philips XL30 FEG ESEM operating at 20.0 Kv. Differential pulse anodic stripping voltammetry (DPASV) and Cyclic voltammetry (CV) were performed in a conventional three-electrode cell with a Autolab-PGATA 302 electrochemical workstation (Metrohm). The conventional three-electrode system included a modified glassy carbon electrode (GCE, 3mm in diameter) as the working electrode, a saturated calomel reference electrode (SCE) and a platinum wire were used as the reference electrode and auxiliary electrode, respectively. DPASV was employed under optimal conditions and involved a preconcentration potential of -1.2 V, a preaccumulation time of 180 s. After the preconcentration, the potential was held at -1.2 V for 30 s under the static condition, followed by a positive-going DVP scan (with a step increment of 5 mV, amplitude of 50mV, and pulse period of 0.2 S) from -1.2 V to 0.2 V. Prior to the next analysis, a preconditioning step (60 s at 0.2 V in stirred solution) was carried out in order to remove residual metal ions on the surface of working electrode.

2.3 Preparation of reduced graphene oxide-chitosan nanocomposite (RGO-CS)

Graphene oxide (GO) was synthesized from graphite powder by a modified Hummers method [38]. The RGO-CS nanocomposites were prepared as follows [39]: 10 mg/mL CS solution was prepared by dissolving CS powder in 0.1 M pH 5.0 sodium acetate solution. 2.5 mg GO was dissolved in 50 mL of doubly distilled water and ultrasonicated for 15 min. Equal volumes of the above CS and GO solutions were mixed with the help of ultrasonic treatment for 5 min in a flask. And then the flask was put into an oil bath for reacting 10 h under 90 °C. During this time, the color of solution in flask was changed from yellow to black gradually until the GO was reduced completely. CS as reductant and stabilizer to transfer GO into RGO.

2.4 Fabrication of RGO-CS/PLL/GCE

The GCE was polished with 1.0 μm , 0.3 μm , and 0.05 μm α -alumina powders and rinsed thoroughly with deionized water between each polishing step and sequentially sonicated in 1:1 HNO_3 , ethanol, and deionized water, and dried at room temperature. Six microliters (0.5 mg/mL) of the RGO-CS nanocomposite suspension was dropped on the surface of GCE and dried in air. The obtained electrode was denoted as RGO-CS/GCE. The electropolymerization of L-lysine on RGO-CS/GCE was performed by dipping the above RGO-CS/GCE into PBS buffer solution (pH 7.5) containing 1.0×10^{-3} mol/L L-lysine by cyclic potential scanning from -1.0 to 0.4 V for 30 cycles with a scan rate of 100 mV s^{-1} .

The final obtained electrode was denoted as RGO-CS/PLL/GCE.

2.5 Procedure for electrochemical testing of Cd(II), Pb(II) and Cu(II)

DPASV was carried out in a 20 ml electrochemical cell containing acetate buffer (0.1 mol L^{-1} , pH 4.5) as supporting electrolyte medium and the target metal ions with varying concentrations from $1.0 \times 10^{-9} \text{ M}$ to $1.0 \times 10^{-8} \text{ M}$. Cd(II), Pb(II) and Cu(II) were deposited at the potential of -1.2 V for 180 s by the reduction of Cd(II), Pb(II) and Cu(II). The anodic stripping (reoxidation of metals to metals ions) of the electrodeposited metals was performed in the potential range of -1.2 to 0.1 V. The peak currents at potential about -0.82 V, -0.54 V and -0.11 V vs. SCE for Cd(II), Pb(II) and Cu(II) were measured, respectively.

3. Results and discussion

3.1 Characterization of RGO-CS

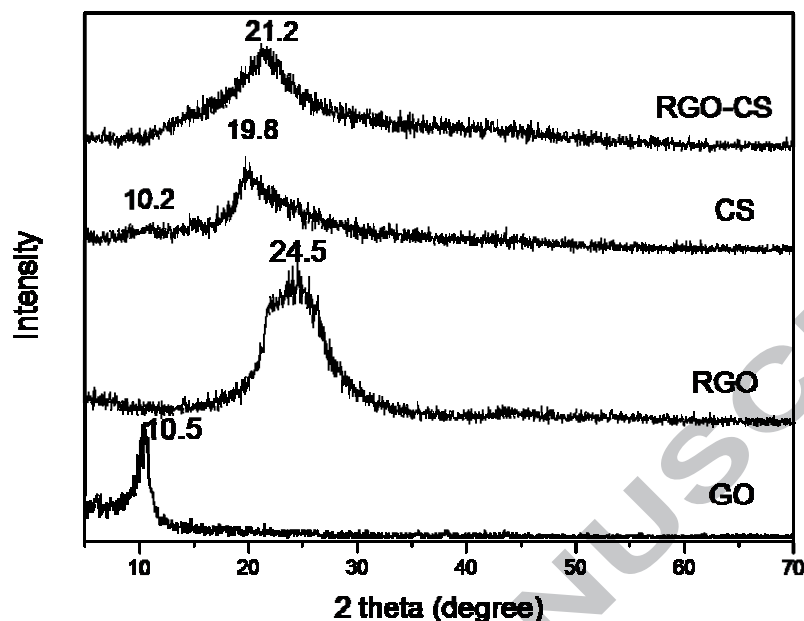


Figure 1 X-ray spectra of GO, RGO, CS and RGO-CS.

Figure 1 shows XRD patterns of GO, RGO, CS and RGO-CS nanocomposite. It can be clearly observed that the sharp XRD peak in GO (d-spacing of 0.84 nm at $2\theta=10.5$) completely disappeared after reduction. Meanwhile a new broad peak appeared in RGO (d-spacing of 0.38 nm at $2\theta=24.5$), which was closer to the typical diffraction peak of graphite, demonstrating that GO had been reduced successfully [40]. The broad peak indicates that RGO possesses a highly disorderly stacked structure [40]. The d-spacing decreased from 0.84 nm to 0.38 nm after the GO was reduced to RGO. This decrease in d-spacing was reasonable because oxygen containing functional groups of GO were disappeared in RGO owing to reduction. The X-ray diffraction pattern shows that CS is a semi-crystalline polymer, and exhibits two main reflection peaks at

$2\theta=10.2$ and 19.8 , which means the existence of crystalline structure and amorphous structure [41]. After the modification of CS to RGO, a diffraction peak appears at about $2\theta=21.2$ corresponding to the d-spacing of 0.42 nm which is larger than that of the pristine graphene nanosheets of 0.38 nm because the interaction of the CS in between graphene layers. It is obvious that after incorporation CS into RGO nanosheets, RGO significantly affects the crystalline structure of CS, as supported by the observation that the characteristic peaks related to the neat CS have greatly altered.

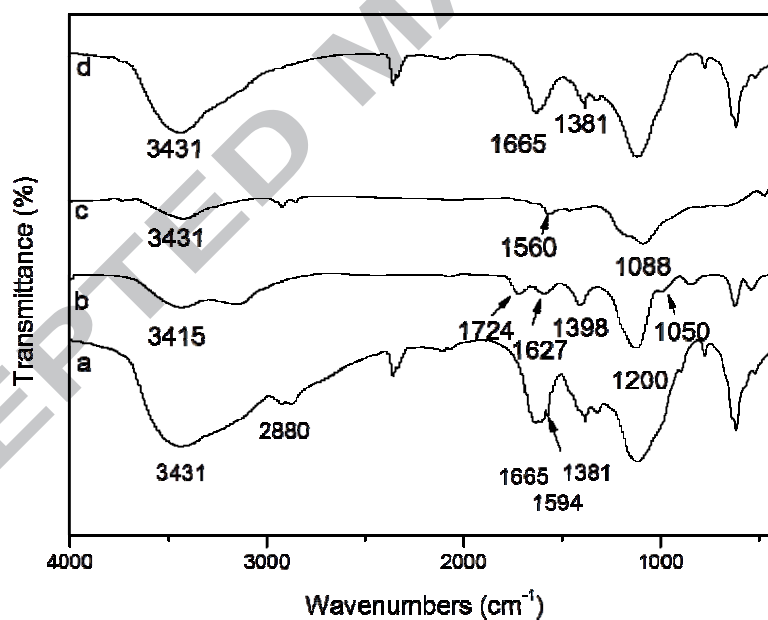


Figure 2 FT-IR spectra of CS (a), GO (b), RGO (c) and RGO-CS (d).

The FT-IR spectra of CS, GO, RGO, and RGO-CS are presented in Figure 2. For the CS (Fig.2 a), the peak at 3431 cm^{-1} was due to the

overlapped of amine and hydroxyl groups stretching vibrations, the appearance of bands at 2880, 1665, 1594 and 1381 cm^{-1} were due to the C-H stretching vibration of CS skeleton, the C-O stretching vibration of -NHCO- and the N-H bending NH_3 [42] and the stretching vibrations of C-N [43], respectively. The FT-IR spectrum of GO (Fig. 2b) exhibits the characteristic absorptions from oxygen-containing functional groups. In detail, the absorption band at 3415 and 1398 cm^{-1} can be assigned to the stretching vibration and deformation vibration of O-H, respectively. The band at 1724 cm^{-1} belongs to the C=O stretching vibration of COOH groups, while the band at 1627 cm^{-1} derives from the vibration of the adsorbed water molecules and/or the contribution of the skeletal vibration of unoxidized graphitic domains. The stretching vibration peaks of C-O (epoxy) and C-O (alkoxy) are observed at 1200 and 1050 cm^{-1} , respectively [44]. As for the RGO (Fig. 2 c), the peak at 3431 cm^{-1} was attributed to O-H stretching vibration, and the peak at 1560 cm^{-1} could correspond to the O-H bending vibration and skeletal ring vibrations, the peak at 1088 cm^{-1} was due to stretching vibrations of C-O-C [45]. Compared with GO, the peaks of oxygen functional groups are disappeared or reduced for RGO, which suggests that many oxygen groups were removed during the transformation from GO to RGO. In comparison with the CS and RGO, in the FT-IR spectra of RGO-CS nanocomposites (Fig. 2 d), C-H stretching vibration of CS skeleton at

2880 cm^{-1} was disappears, which could be ascribed to the hydrogen bonds occurred between CS and RGO, and electrostatic interaction between polycationic in CS and negative charge in RGO. This demonstrated that RGO-CS nanocomposites were successfully synthesized.

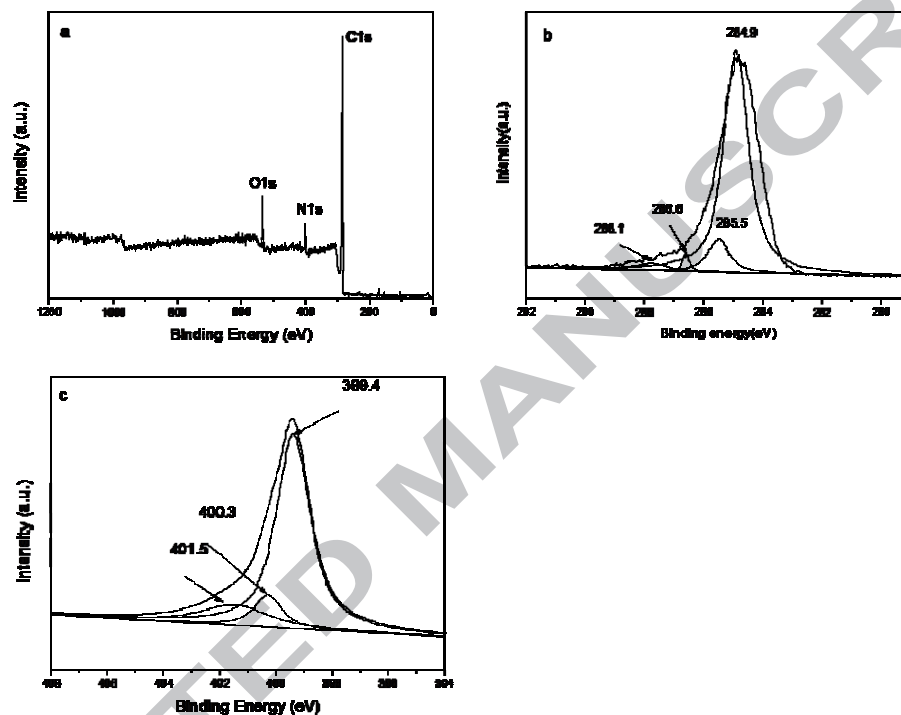


Figure 3 XPS spectrum of RGO-CS nanocomposite: wide scan (a); C 1s (b) and N 1s (c)

The chemical composition and electronic structure of RGO-CS were characterized by XPS measurements (Figure 3). The wide-scan XPS spectrum of RGO-CS (Fig. 3a) shows three peaks at 284.7 eV, 399.7 eV and 531.2 eV corresponding to C 1s, N 1s and O 1s respectively, indicating the existence of C, N and O element. The main C 1s peak in Fig. 3b is dominated by elemental carbon at 284.9 eV, attributing mainly to sp^2 hybridized carbon. The weak C 1s peak at 285.5 eV is attributed

mainly to the oxygen bound species C-O, and two peaks at 286.6 and 288.1 eV are assigned to the oxygen bound species C=O and O-C=O, respectively [40, 46]. After reduction of GO, C-O, C=O and O-C=O binding energy decreased significantly. According to the XPS peak area, the O/C ratio of graphene is 6.96%, far less than the O/C ratio of graphene oxide by 45.39%. This suggests that the O elements have been reduced successfully in the graphene. Meanwhile, from N 1s spectra of RGO-CS (Figure 3 c), it is obviously observed that three peaks centered at 399.4, 400.3 and 401.5 eV, which correspond to the amine (C-NH₂), amide (C-NH-C=O) and the protonated amine species (C-N⁺) [47, 48]. The result was confirmed that the CS has anchored into the RGO composite.

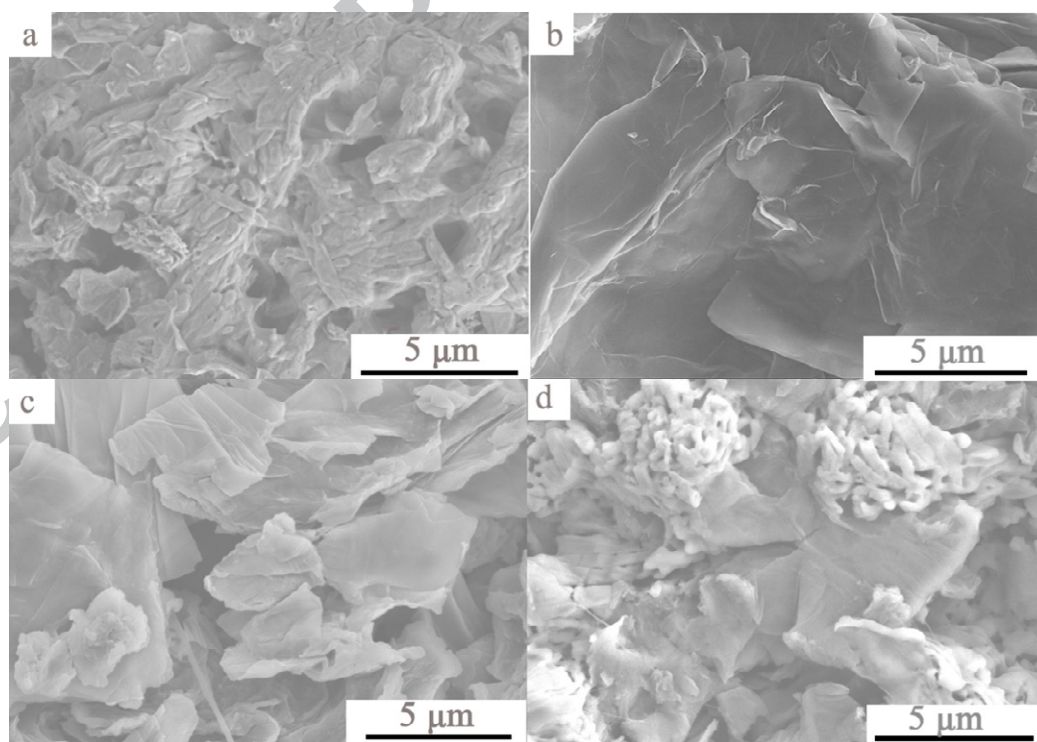


Figure 4 SEM images of CS (a), RGO (b), RGO-CS (c) and RGO-CS/PLL (d).

The morphologies of CS, RGO, RGO-CS and RGO-CS/PLL were characterized by SEM as shown in Fig. 4. For CS (Fig. 4a), there is a rough surface and the internal structure is solid. It is shown in Fig. 4b that the RGO sheets displayed a typical wrinkled and crumpled smooth surface of graphene, and the sheets stacked together to form a typical multi-layer structure. For RGO-CS (Fig.4c), the SEM image reveals an entirely different surface morphology compared to the CS. It can be seen that CS was fully encapsulated in RGO sheets or as a spacer to inhibit of their aggregation. Meanwhile, some micro-wrinkled stripes protuberances can be found in the RGO-CS surface, which can provide the interspace to adsorption of the heavy metal ions. For RGO-CS/ PLL (Fig.4d), a large number of rod-like fibre closely attach on the surface of RGO-CS after the electropolymerization of PLL, indicating the formation of PLL onto RGO-CS sheets.

3.2. The electropolymerization of L-lysine

The L-lysine polymer film modified electrode was synthesized by electropolymerization of L-lysine on the RGO-CS/GCE surface. The experiment was performed in PBS buffer solution (pH 7.5) containing 1.0×10^{-3} mol/L L-lysine with cyclic voltammetric sweeps in the potential range from -1.0 to $+2.0$ V. As shown in Fig. 5, the cathodic peak is observed at -0.6 V on continuous scanning, reflecting the continuous

growth of the PLL film on the RGO-CS/GCE surface. After 30 cycles, the electropolymerization procedure was completed, and electrode was washed with distilled water to remove any physically adsorbed material. After dried in air, a blue film (PLL film) could be observed on the surface of electrode. At a higher positive potential, L-lysine monomer can be oxidized to form α -amino free radical, and the free radical can be linked to electrode surface [49]. The scan cycles and the monomer concentration can affect the electropolymerization process and the thickness of films. The results showed that stable polymer film was achieved under a scan cycle of 30 with the monomer concentration of L-lysine in PBS buffer solution 1.0×10^{-3} mol/L.

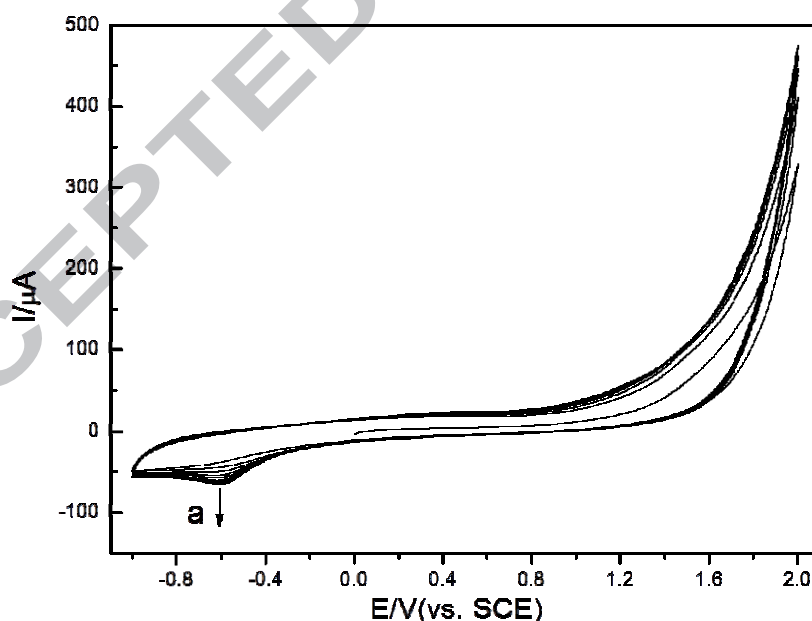


Fig. 5. Repetitive cyclic voltammograms of 1.0×10^{-3} mol/L L-lysine in PBS solution (pH 7.5) at the RGO-CS/GCE. Scan rate: 100 mV/s.

3.3 Electrochemical characterization of the modified electrodes

The electrochemical performance of modified GCE was compared with bare GCE in 1.0 mmol/L $\text{Fe}(\text{CN})_6^{3-/4-}$ and 0.5 mol/L KCl mixture solution, and the cyclic voltammograms were shown in Fig. 6.

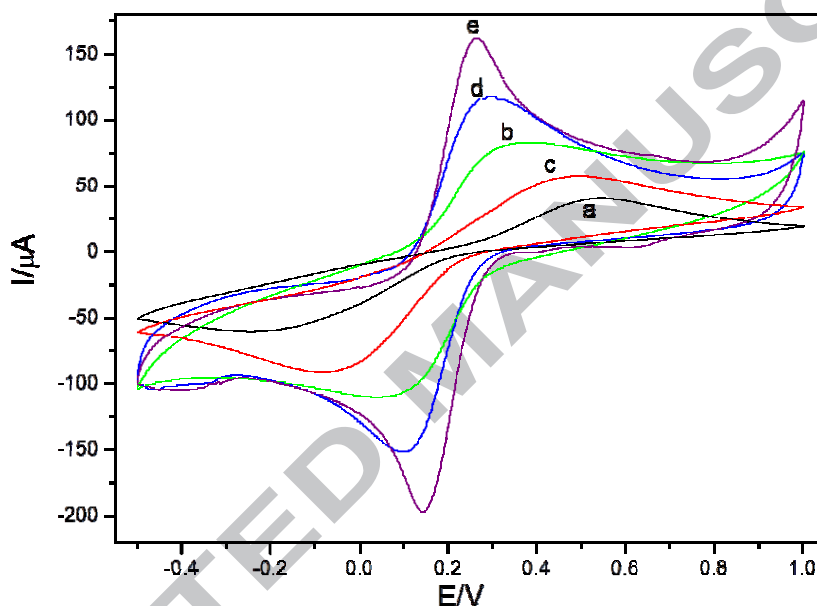


Figure 6 CVs of bare GCE (a), RGO/GCE (b), RGO-CS/GCE (c), RGO/PLL/GCE (d) and RGO-CS/PLL/GCE (e) in 1.0 mmol/L $[\text{Fe}(\text{CN})_6]^{3-/4-}$ solution containing 0.5 mol/L KCl.

Figure 6 shows the cyclic voltammograms (CVs) for different electrodes in 1.0 mmol/L $\text{K}_3\text{Fe}[\text{CN}]_6$ and 0.5 mol/L KCl mixture solution. As shown in Fig. 6, the CV of the bare GCE (curve a) showed a pair of well-defined quasi-reversible peaks, and the peak-to-peak separation (ΔE_p) was calculated as 0.747 V. On RGO/GCE (curve b), the redox peak currents also increased with the ΔE_p value as 0.221 V, which

could be attributed to the presence of RGO film on the electrode surface with excellent electrical conductivity and fast electrode transfer rate. For RGO-CS/GCE electrode (curve c), the ΔE_p increased and peak current (I_p) decreased, indicating that CS could block the electron transfer between electrode surface and $[\text{Fe}(\text{CN})_6]^{3-/4-}$. That is attributed to the insulativity of CS. On RGO/PLL/GCE (curve d), the peak currents of redox peaks increased with the ΔE_p value decreased, which could be attributed to attraction of the amino group cations of the PLL to $\text{Fe}(\text{CN})_6^{3-/4-}$ leading to more easily redox of $\text{Fe}(\text{CN})_6^{3-/4-}$ [49]. On RGO-CS/PLL/GCE (curve e), the biggest redox peak currents appeared with the smallest ΔE_p value of 112 mV. Hydrogen bonds exist between CS and PLL. The rate of electron transfer is accelerated by the movement of electrons through hydrogen bonds between CS and PLL [50]. According to Randles-Sevcik equation [51], $i_{pa} = 2.69 \times 10^5 n^{3/2} A C D^{1/2} v^{1/2}$, where i_{pa} is the anodic peak current (μA), n is the number of electrons transferred, A is electroactive surface area (cm^2), C is concentration (mol/cm^3), v is the potential scanning rate (mV s^{-1}) and D is the diffusion coefficient (cm^2/s). The electroactive surface area is 0.093, 0.086, 0.074, 0.069 and 0.052 cm^2 for RGO-CS/PLL/GCE, RGO/PLL/GCE, RGO-CS/GCE, RGO/GCE and bare GCE, respectively. This indicates that the electrochemical reactivity of RGO-CS/PLL/GCE electrode is better than that of RGO/PLL/GCE, RGO-CS/GCE, RGO/GCE and bare GCE electrode.

Electrochemical impedance spectrum (EIS) is also used for investigation of the capability of electron transfer on the surface of different electrodes. In electrochemical impedance measurement, the semicircle diameter of impedance equals the electron transfer resistance (R_{et}), which controls the electron transfer kinetics of the redox probe at the electrode surface. Fig. 7 is the Nyquist diagrams of bare GCE (a), RGO/GCE (b), RGO-CS/GCE(c), RGO/PLL/GCE(d) and RGO-CS/PLL/GCE (e) recorded in 1.0 mmol/L $\text{Fe}(\text{CN})_6^{3-/4-}$ solution containing 0.5 mol/L KCl with the frequency swept from 10^5 to 0.1 Hz. A semicircle of about 341 Ω in diameter for bare (curve a) at higher frequencies indicates a diffusion limited step of the electrochemical process. After modified by RGO (curve b), the semicircle diameter was remarkably decreased to about 197 Ω , which showed that the presence of RGO could effectively accelerate the electron transfer. But, after modified RGO-CS nanocomposites (curve c), the semicircle diameter increased significantly to about 264 Ω compared with the RGO/GCE. The results demonstrated RGO-CS nanomaterials could increase the electron transfer resistance to the flow of electrons and decrease the conductivity of the electrode, which is consistent with the CV results above. This also demonstrated that RGO-CS film was successfully immobilized on the GCE surface. After modified by RGO/PLL and RGO-CS/PLL (curve d and e), the semicircle diameter was decreased to about 179 and 83 Ω . The

curve of RGO/PLL/GCE and RGO-CS/PLL/GCE exhibited gradually smaller radius of semicircles, which showed that assembled PLL molecules with positive charges are able to attract $[\text{Fe}(\text{CN})_6]^{3-/4-}$ electrostatically due to containing plentiful active amino groups [35]. Moreover, the rate of electron transfer is accelerated by the movement of electrons through hydrogen bonds between CS and PLL.

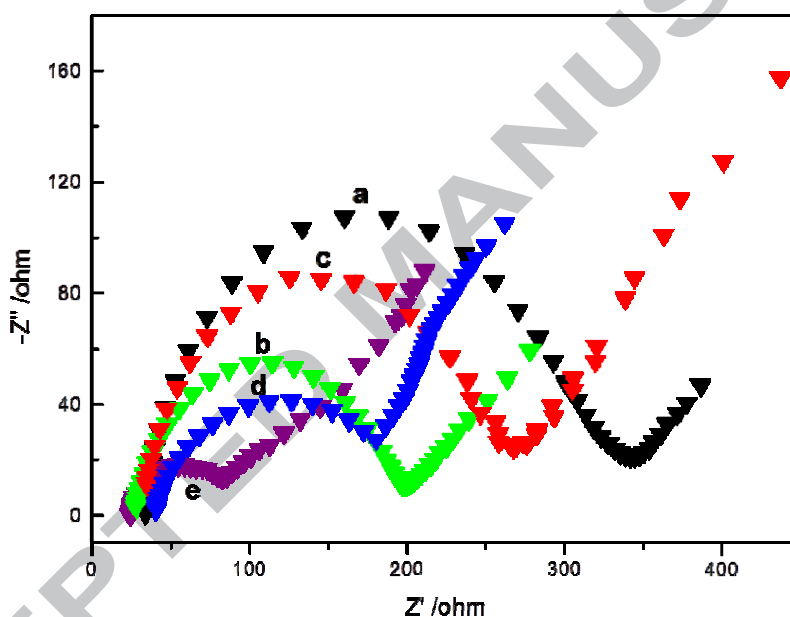


Figure 7 Electrochemical impedance spectra of bare GCE (a), RGO/GCE (b), RGO-CS/GCE (c), RGO/PLL/GCE (d) and RGO-CS/PLL/GCE (e) in 1.0 mmol/L $[\text{Fe}(\text{CN})_6]^{3-/4-}$ solution containing 0.1 mol/L KCl.

3.4 Stripping behavior for heavy metal ions

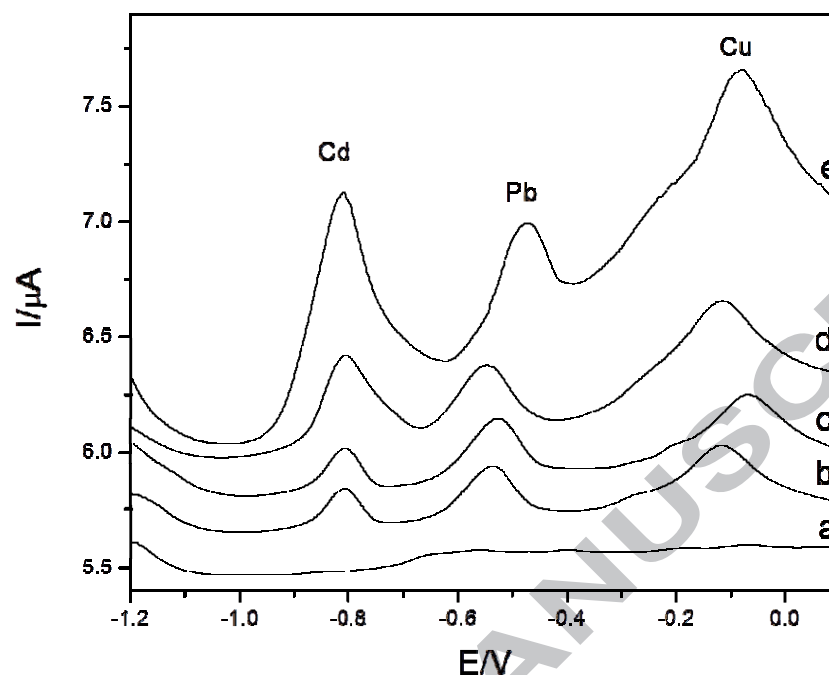
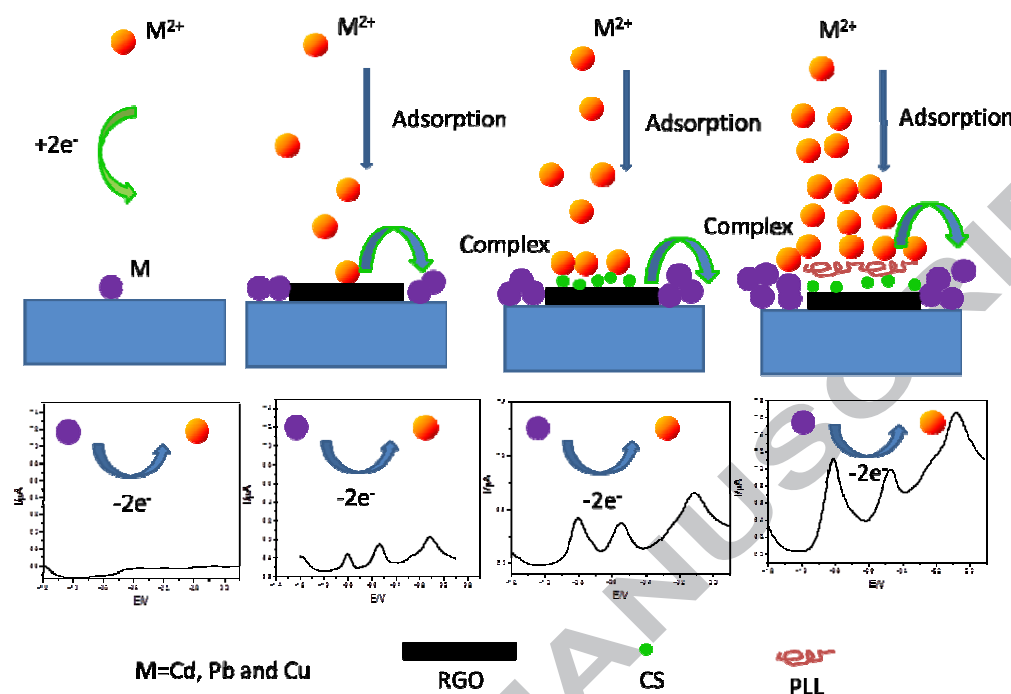


Figure 8 DPASV stripping signals for $1.0 \mu\text{g L}^{-1}$ each of Cd(II), Pb(II) and Cu(II) on bare GCE (a), RGO/GCE (b), RGO-CS/GCE(c), RGO/PLL/GCE (d) and RGO-CS/PLL/GCE (e) in 0.1 M acetate buffer (pH 4.5). Conditions: deposition potential: -1.20 V vs. SCE; deposition time: 180s; resting time: 30s.

Fig. 8 shows the DPASV curves of Cd (II), Pb (II) and Cu (II) at different modified electrodes in 0.1M pH 4.5 acetate buffer solution. For the bare GCE (curve a), the stripping peaks for metal ions were overlapped in the potential range -1.2-0.2 V vs. SCE, so that the simultaneous determination of these heavy metal ions is impossible. For RGO/GCE, RGO-CS/GCE and RGO/PLL/GCE (curve b, c and d), it can be seen obvious increase in their stripping peak currents. For RGO-CS/PLL/GCE (curve e), the peak potential of three stripping peaks are larger than 200 mV. The enhanced stripping current for

RGO-CS/PLL/GCE is due to that larger amount of reduced metal could be deposited at deposition potential by the reduction of metal ions. RGO-CS/PLL/GCE can provide excellent electrochemical platforms for target heavy metals sensing due to the combining of enlarged active surface area, strong adsorption capability and the specific complexing ability of the CS and PLL in analysis of environmental samples. These advantages make the RGO-CS/PLL nanocomposites be an ideal sensing nanomaterial suitable for the electrochemical detection or preconcentration of trace amounts of heavy metal ions, as illustrated by Scheme 1. It is known that peak response of heavy metal ion is directly proportional to its concentration in solution. At this stage, large amounts of target metal ions (Cd (II), Pb (II) and Cu (II)) can be adsorbed or forming a complex onto the surfaces of the nanomaterials. When the potential is held at deposition region, Cd (II), Pb (II) and Cu (II) will be reduced on the surface of the electrode and metal crystal growth occurs. In this way, Cd (II), Pb (II) and Cu (II) are accumulated and then stripped out. The more Cd (II), Pb (II) and Cu (II) are adsorbed on the surface of RGO-CS/PLL nanocomposite, the more it will be released, thus strengthening the stripping peak response. Moreover, the potentials for the reoxidation of Cd (II), Pb (II) and Cu (II) on RGO, RGO-CS, RGO/PLL are more negative than that observed on RGO-CS/PLL modified GCE, indicating that the reoxidation process is more favorable

at the RGO-CS/PLL surface.



Scheme 1 Schematic diagram of RGO-CS/PLL for electrochemical detection of heavy metal ions.

3.5 Optimization of experimental conditions

3.5.1 Effect of supporting electrolytes

In order to gain high sensitive electrochemical performance with RGO-CS/PLL/GCE to heavy metal ions, the effect of experimental parameters (supporting electrolytes, pH, deposition time and deposition potential) were discussed.

Since buffer solution had the ability to fix pH value and ionic strength, therefore, it was widely employed as supporting electrolyte in the practical analysis. Fig. 9 shows the stripping voltammograms in 0.1 M pH 4.5 solutions of PBS, boric acid buffer solution and NaAc-HAc to

response toward Cd(II), Pb(II) and Cu(II) on the RGO-CS/PLL/GCE. No obvious peak current were observed in PBS and boric acid buffer solution. NaAc-HAc has the optimum sensitivity towards heavy metal ions because the stripping peaks were observed.

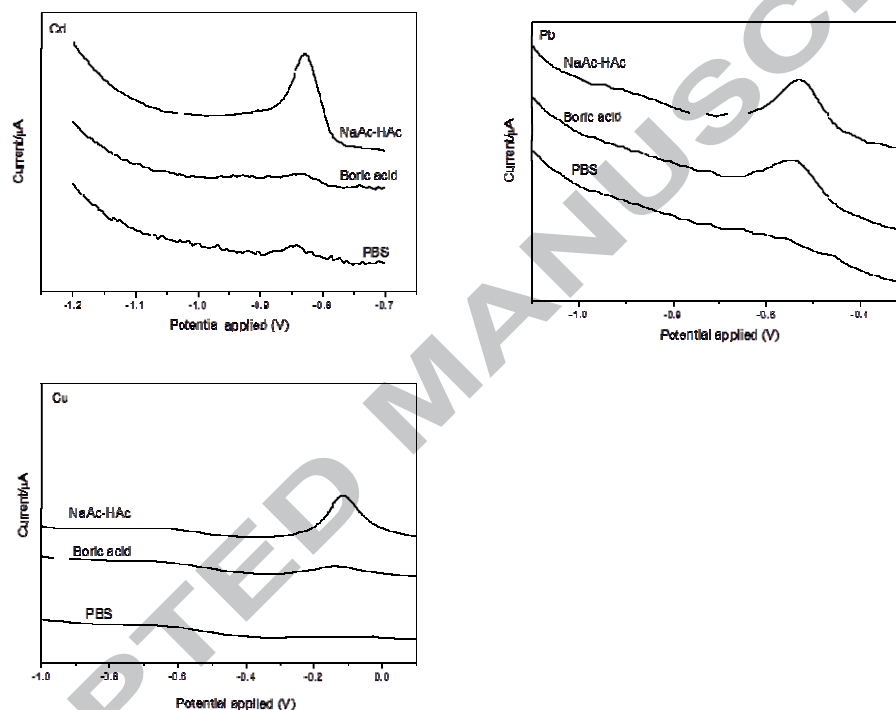


Fig. 9 Effect of supporting electrolytes on the stripping peak currents for $1.0 \mu\text{g L}^{-1}$ each of Cd(II), Pb(II) and Cu(II) solution. DPASV conditions are identical to those in Fig. 8.

3.5.2 Effect of pH

The influence of pH value on the stripping peak currents was studied in 0.1M NaAc-HAc buffer with the pH range between 3.5 and 5.5 in the DPASV and the result is shown in Figure 10 (a). The pH value influence the stripping peak currents and the maximum DPASV peak was observed

at the pH 4.5. The lower pH value (3.5 and 4.0) result in a reduction peak current, which is possible due to the protonation of the hydrophilic groups reducing the absorption of metal ions[19]. The peak current at the higher pH value (5.0 and 5.5) decreased, which is possible due to the hydrolysis of Cd(II), Pb(II) and Cu(II). So, pH 4.5 was chosen as the optimal pH value for the analysis.

3.5.3 Effect of deposition potential

Deposition potential is an important parameter for stripping techniques and has a significant effect on the sensitivity. The influence of deposition potential on the stripping response of Cd(II), Pb(II) and Cu(II) were studied. The deposition potential was varied between -0.9 and -1.4 V under optimized pH condition. The results presented in Fig. 10 (b) indicates that the peak currents of mixed ions increase remarkably as the deposition potential negatively shifts from -0.9 to -1.2 V. The more negative the deposition potential, the more easily Cd(II), Pb(II) and Cu(II) were reduced, thereby causing the obvious increase of the stripping peak currents. Three well-separated stripping peaks corresponding to Cd(II), Pb(II) and Cu(II) are obtained at -1.2 V with the relative higher peak currents. When the deposition potential becoming more negative, the reproducibility of stripping currents for Cd(II), Pb(II) and Cu(II) became poor, because hydrogen evolution was beginning to be significant in the medium at such negative potentials [19]. Furthermore, the metals

deposited on the electrode surface might be damaged by the hydrogen bubble and lead to decrease in current signals at very negative potentials [52]. Therefore, in order to obtain the high sensitivities of three ions, the deposition potential of -1.2 V was chosen for the further work.

3.5.4 Effect of deposition time

Fig. 10 (c) shows a plot of peak current versus deposition time. As can be seen, the peak currents are almost linearly proportional to deposition time, and this may be attributed to the fact that the amount of metal ions at the modified electrode surface greatly increases due to electrochemical deposition. Here the choice of 180 s was a compromise between reducing practical measurement time and enhancing high sensitivity.

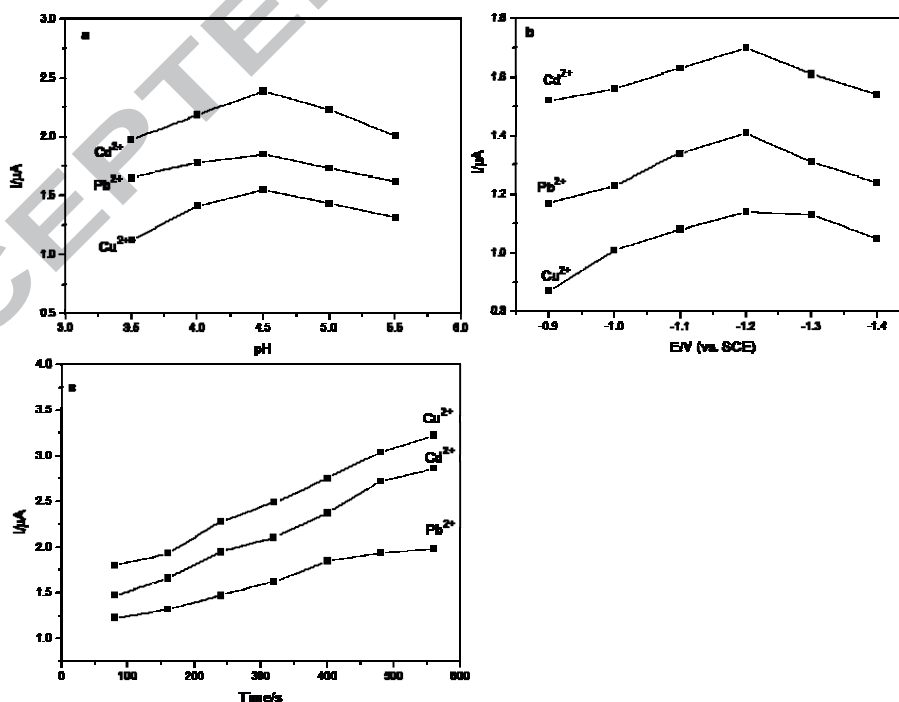


Figure 10 The effects of solution pH (a), deposition potential (b), and deposition time (c) on the stripping currents of Cd(II), Pb(II) and Cu(II) at RGO-CS/PLL/GCE electrode in 0.1 M acetate buffer solution containing $1.0 \mu\text{g L}^{-1}$ of Cd(II), Pb(II) and Cu(II).

3.6. Electrochemical detection of Cd(II), Pb(II) and Cu(II)

The DPASV characterization was further performed to investigate simultaneous determination of Cd(II), Pb(II) and Cu(II). For different concentrations of Cd(II), Pb(II) and Cu(II) in the mixture was performed at the RGO-CS/PLL/GCE and experiment was investigated when changing the concentration of one species whereas the others were kept constant. Fig. 11 shows the DVASV responses of the RGO-CS/PLL/GCE towards Cd(II), Pb(II) and Cu(II) at different concentrations. Cd(II), Pb(II) and Cu(II) are detected at potentials of -0.82, -0.54 and -0.11V, respectively. The corresponding calibration curve from 0.05 to 10.0 $\mu\text{g.L}^{-1}$ were

$$i_p(\mu\text{A})=6.91+2.51 C (\mu\text{g.L}^{-1}); (0.05 \text{ to } 2.0 \mu\text{g.L}^{-1}, R=0.996) \quad (1)$$

$$i_p(\mu\text{A})=8.26+0.57C (\mu\text{g.L}^{-1}); (2.0 \text{ to } 10.0 \mu\text{g.L}^{-1}, R=0.998) \quad (2)$$

for Cd(II),

$$i_p(\mu\text{A})=7.01+0.45C(\mu\text{g.L}^{-1}); (0.05 \text{ to } 2.0 \mu\text{g.L}^{-1}, R=0.996) \quad (3)$$

$$i_p(\mu\text{A})=7.41+0.09 C(\mu\text{g.L}^{-1}); (2.0 \text{ to } 10.0 \mu\text{g.L}^{-1}, R=0.999) \quad (4)$$

for Pb(II),

and

$$i_p(\mu\text{A})=7.20+0.51 C(\mu\text{g.L}^{-1}); (0.05 \mu\text{g.L}^{-1} \text{ to } 2.0 \mu\text{g.L}^{-1}, R=0.997) \quad (5)$$

$$i_p(\mu A)=7.62+0.168 C(\mu g.L^{-1}); (2.0 \mu g.L^{-1} \text{ to } 10.0 \mu g.L^{-1}, R=0.998) \quad (6)$$

for Cu(II), respectively. The limit of detection (LOD) (based on a signal-to-noise ratio of 3 ($S/N = 3$)) were found to be $0.01 \mu g.L^{-1}$ for Cd(II), $0.02 \mu g.L^{-1}$ for Pb(II) and $0.02 \mu g.L^{-1}$ for Cu(II), respectively. Moreover, the limits of detection for Cd(II), Pb(II) and Cu(II) are much lower than the guideline values in drinking water given by the WHO (For Cd(II): 0.003 mg.L^{-1} ; Pb(II): 0.01 mg.L^{-1} ; Cu(II): 2.0 mg.L^{-1}). It is clear that the RGO-CS/PLL electrode shows the lower limit of detection than most of the electrodes listed in Table 1.

In addition, we measured the maximum adsorption capacity toward Cd(II), Pb(II) and Cu(II) onto RGO-CS/PLL at room temperature, as shown in Fig. 11d. It can be seen that the RGO-CS/PLL could adsorb these ions and the maximum adsorption capacity toward Cd(II), Pb(II) and Cu(II) is 47.14 , 18.96 and 16.33 mg.g^{-1} , respectively. This experiment leads to a conclusion that the sensitivity is related to the adsorption capacity of the materials.

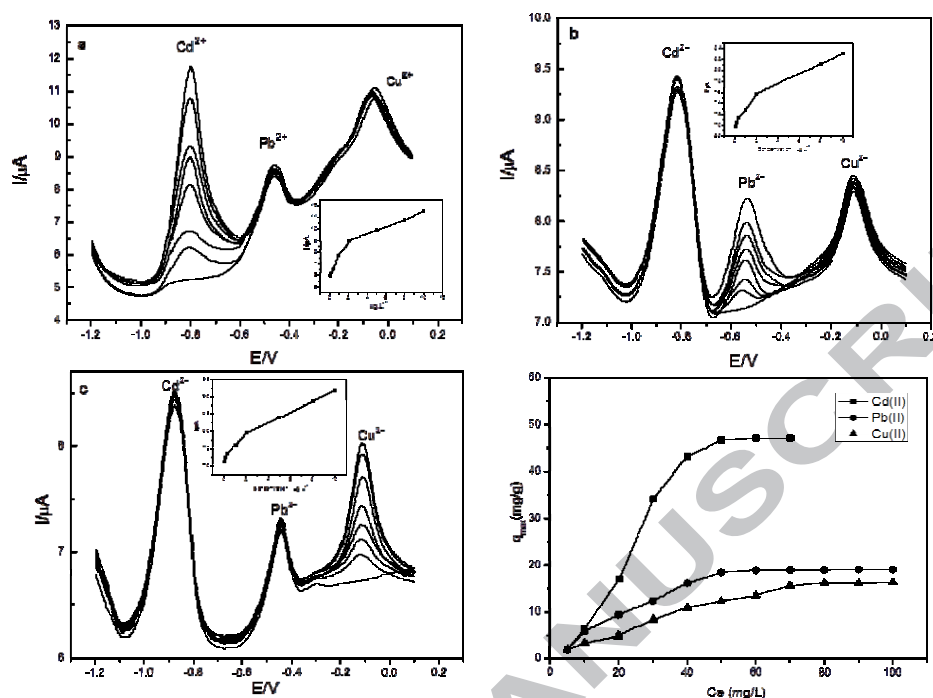


Figure 11 DPASV of heavy metal ions at RGO-CS/PLL/GCE (a) $8.0 \mu g.L^{-1}$ Pb(II) and $8.0 \mu g.L^{-1}$ Cu(II) with concentrations varying in the range of $0-10.0 \mu g.L^{-1}$ Cd(II); (b) $8.0 \mu g.L^{-1}$ Cd(II) and $8.0 \mu g.L^{-1}$ Cu(II) with concentrations varying in the range of $0-10.0 \mu g.L^{-1}$ Pb(II); (c) $8.0 \mu g.L^{-1}$ Cd(II) and $8.0 \mu g.L^{-1}$ Pb(II) with concentrations varying in the range of $0-10.0 \mu g.L^{-1}$ Cu(II). Supporting electrolyte: 0.1 M acetate buffer (pH 4.5); deposition potential: -1.2V; deposition time: 180s; quiet time: 10s. (d) Adsorption isotherm of RGO-CS/PLL for Cd(II), Pb(II) and Cu(II). Adsorbent dose= $1 g.L^{-1}$, contact time=24 h, $T=298K$.

Table 1 Comparison of some characteristics of the different modified electrodes for the determination of Cd(II), Pb(II) and Cu(II)

Electrode Materials	Detection limit ($\mu\text{g.L}^{-1}$)			Linear range ($\mu\text{g.L}^{-1}$)			Reference
	Cd	Pb	Cu	Cd	Pb	Cu	
RGO-CS/PLL/GCE	0.01	0.02	0.02	0.05-10.0	0.05-10.0	0.05-10.0	This work
MoS ₂ -xI _x NWs/GCE ^a	0.10	0.45	0.20	0.5-150	1.5-450	0.8-240	[53]
ZYMCPE ^b	1.0	3.6	---	5.62-112.40	5.18-20.72	---	[54]
Nafion/G/PANI ^c	0.1	0.1	---	1-300	1-300	---	[15]
RGO/Bi/CPE ^d	2.8	0.55	26	20.0-120	20.0-120	20.0-100	[55]
BTSBA/GPE ^e	1.12	0.062	0.32	90.0	2.10	6.40	[56]
G/MWCNTs/Bi/GCE ^f	0.1	0.2	---	0.5-30	0.5-30	---	[19]
GO/AP/GCE ^g	0.0004	---	0.0002	0.001-0.056	---	0.0006-0.032-	[5]
PPh ₃ /MWCNT/IL/CPE ^h	8.32	12.43	---	0.011-16.86	0.021-31.08	---	[52]
Bi/Cu/SPE ⁱ	59.6	172.0	---	111.3-1348	269.4-2693	---	[57]
Bi/Au-GN-Cys/GCE ^j	0.1	0.05	---	0.500-40.0	0.500-40.0	---	[58]
SbNP/MWCNT/CPE ^k	0.77	0.65	---	10.0-60.0	10.0-60.0	---	[59]

a: MoS₂-xI_x nanowires modified GCE;

b: Zeolite NH₄-Y modified carbon paste electrode;

c: Nafion/ Graphene/Polyaniline nanocomposites;

d: Reduced graphene oxide(RGO)/Bismuth (B)/carbon paste electrode;

e: Graphite-polyurethane composite electrode with SBA-15 silica organofunctionalized with 2-benzothiazolethiol;

f: Graphene/MWCNTs/Bi/GCE;

g: Graphene oxide terminated p-aminophenyl modified GCE;

h: Triphenylphosphine/MWCNTs/Ionic liquid/carbon paste electrode;

i: Bi/Cu/Screen-printed electrodes

j: Gold nanoparticle-graphene-cysteine composite modified bismuth film electrode

k: Antimony nanoparticle-multiwalled carbon nanotubes composite immobilized at carbon paste electrode

3.7 Simultaneous determination of Cd(II), Pb(II) and Cu(II)

Then under the optimal experiment conditions, the behaviors of DPASV analysis for the simultaneous determination of Cd(II), Pb(II) and Cu(II) was shown in Figure 12. It is seen from Fig. 12 that the voltammetric peak for the stripping of Cd(II), Pb(II) and Cu(II) in the RGO-CS/PLL/GCE appears at different potentials with a separation of

270 mV-430 mV between the stripping peaks, such a separation is large enough, and the simultaneous detection using RGO-CS/PLL/GCE is feasible. Linear increase in the peak current was obtained while increasing the concentration of Cd(II), Pb(II) and Cu(II) ions simultaneously.

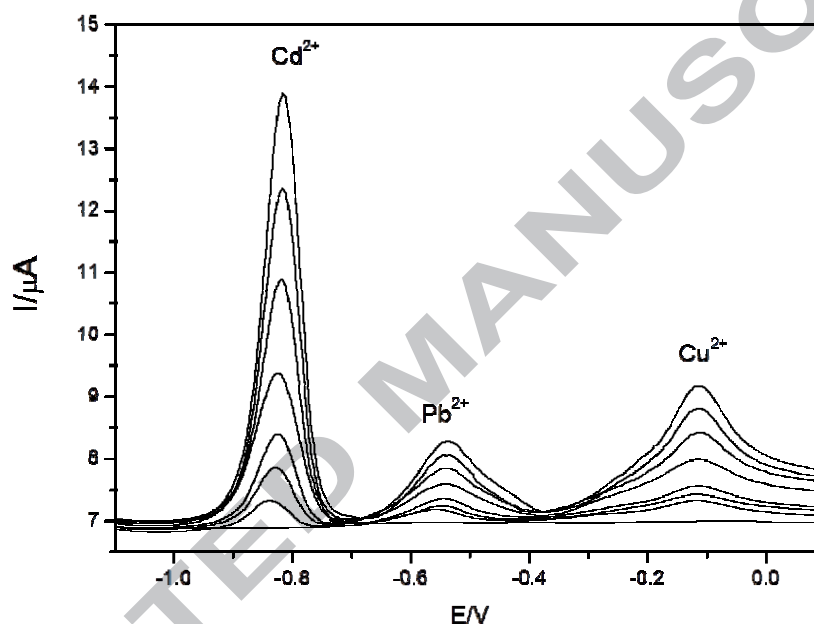


Figure 12 DPASV of the RGO-CS/PLL/GCE for the simultaneous detection of Cd(II), Pb(II) and Cu(II) mixed ions in 0.1 M acetate buffer (pH 4.5) under the optimal experimental conditions

3.8. Interference study for Cd(II), Pb(II) and Cu(II) detection

The presence of non-target metals in solution can interfere with DPASV determination of heavy metals, so the tolerance ratio for common interfering metals (Zn(II), Mn(II), Ni(II), Fe(II), Fe(III), and Co(II), Mg(II), Ba(II), Ca(II), Al(III), K(I), Ag(I), Na(I)) was investigated. The concentration of each species tested is provided in Table 2. A metal

concentration altering an analyte peak current by >5% was considered inhibitory. By limiting the scanning potential in the system from -1.6 to -0.5 V, we do not observe appreciable interference from metals at the concentrations listed in Table 2. The RGO-CS/PLL/GCE nanocomposite modified electrode was not affected by other metal species, demonstrating the selectivity of this electrode for Cd(II), Pb(II) and Cu(II).

Table 2. Tolerance ratio of interfering metal ions in the electrochemical determination of $1.0 \mu\text{g L}^{-1}$ of Cd(II), Pb(II) and Cu(II) on RGO-CS/PLL/GCE electrode

Interfering Metal	5 % Tolerance Ratio		
	Cd	Pb	Cu
Zn(II)	30.0	30.0	30.0
Mn(II)	50.0	50.0	50.0
Ni(II)	25.0	25.0	25.0
Fe(II)	30.0	30.0	30.0
Fe(III)	30.0	30.0	30.0
Co(II)	30.0	30.0	50.0
Mg(II)	110.0	110.0	110.0
Ba(II)	100.0	100.0	100.0
Ca(II)	150.0	150.0	150.0
Al(III)	150.0	150.0	150.0
K(I)	1000.0	1000.0	1000.0
Ag(I)	700.0	700.0	700.0
Na(I)	1000.0	1000.0	1000.0

3.9 Reproducibility and stability of electrode

The repeatability and stability of the electrode could be evaluated by a series of repetitive experiment of DPASV. Firstly, the repeatability was examined by performing 10 times at the same standard solutions ($5.0 \mu\text{g}$

L^{-1}) Cd(II), Pb(II) and Cu(II), the relative standard deviation were 1.7%, 2.2% and 2.4% of each of Cd(II), Pb(II) and Cu(II), which showing that the reproducibility were good. At the same time, the stability was examined by recorded every day over 3 weeks, the maximum deviations obtained were 4.2%, 3.9% and 4.4%, which showing that the composites has long-time stability.

3.10 Analysis of real samples

In order to evaluate the applicability of this method, the proposed electrode was used to examined the concentration of Cd(II), Pb(II) and Cu(II) in tap water by the standard addition method, and the same as above mentioned optimal conditions was used in this experiments. The result listed in Table 3, the recovery of Cd(II) were 96.00-103.33%, and the recovery of Pb(II) were 93.33-97.6%, and 96.66-101.33 of Cu(II). Above all, the RGO-CS/GCE/GCE can be utilized as a selective and precise electrode to determination metal ions, and have a better reproducible.

Table 3 Recovery result for Cd(II), Pb(II) and Cu(II) in tap water.

Heavy metal ion	Added ($\mu\text{g L}^{-1}$)	Found ($\mu\text{g L}^{-1}$)	Recovery (%)
Cd(II)	0.30	0.31	103.33
	1.50	1.45	96.67
	5.0	4.80	96.00
Pb(II)	0.30	0.28	93.33
	1.50	1.43	95.33
	5.0	4.88	97.6
Cu(II)	0.30	0.29	96.66
	1.50	1.52	101.33
	5.0	4.84	96.80

4. Conclusions

In this paper, a new enhanced sensing platform base on the RGO-CS/PLL modified GCE was established. The combination of RGO-CS with PLL can provide excellent electrochemical platforms for target heavy metals sensing due to the combining of enlarged active surface area, strong adsorption capability and the specific complexing ability of the CS and PLL in analysis of environmental samples. This working electrode was used to simultaneously determination of Cd(II), Pb(II) and Cu(II) heavy metal ions. Three good defined and sharp stripping peaks can be obtained for Cd(II), Pb(II) and Cu(II), and the stripping peak currents show a good linear proportionality from 0.05 $\mu\text{g.L}^{-1}$ to 10.0 $\mu\text{g.L}^{-1}$ with a detection limit 0.01 $\mu\text{g.L}^{-1}$ for Cd(II), 0.02 $\mu\text{g.L}^{-1}$ for Pb(II) and 0.02 $\mu\text{g.L}^{-1}$ for Cu(II), respectively. The limits of detection for Cd(II), Pb(II) and Cu(II) are much lower than the guideline values in drinking water given by the WHO. Finally, the proposed electrode was successful applied to the analysis of the tap water samples.

Acknowledgment

The authors gratefully acknowledge financial support from the Liaoning Natural Science Foundation (2015020197) and the National Natural Science Foundation of China (Grant no. 51403130).

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Graphical abstract

